Companion XXXIX: The Thermal Sunyaev–Zel'dovich Effect in the Relaxation-Driven Cyclic Cosmology

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Abstract

The thermal Sunyaev–Zel'dovich (tSZ) effect provides a direct probe of the pressure distribution in galaxy clusters and the thermodynamic state of the intracluster medium (ICM). In the Relaxation-Driven Cyclic Cosmology (RDCC), the scale-dependent evolution of the gravitational potential modifies the formation history of clusters, the temperature and density structure of their gas, and the coherence of the surrounding large-scale environment. This Companion develops a continuous description of the tSZ effect in RDCC, deriving how the relaxation scale influences the Compton-y parameter, the pressure profile, and the integrated tSZ signal. The resulting framework provides a natural interpretation of observed deviations from self-similar scaling relations and offers new predictions for upcoming CMB surveys.

1 Introduction

The thermal Sunyaev–Zel'dovich effect arises from the inverse Compton scattering of cosmic microwave background photons by hot electrons in galaxy clusters. It is a unique probe of the thermal energy content of the intracluster medium and is largely insensitive to redshift. In the standard cosmological model, the tSZ signal is determined by the depth of the gravitational potential, the virial temperature of the gas, and the pressure profile of the cluster.

In RDCC, the gravitational potential evolves differently. The relaxation mechanism suppresses the growth of small-scale modes and enhances the coherence of large-scale modes. This modifies the collapse history of clusters, the thermodynamic structure of their gas, and the pressure distribution that determines the tSZ signal. The tSZ effect becomes a direct tracer of the relaxation scale k_{rel} and the temporal structure of the gravitational field.

2 The RDCC Potential and Cluster Formation

The formation of galaxy clusters is governed by the evolution of the gravitational potential. In RDCC, this potential takes the form

$$\Phi_{\mathrm{RDCC}}(k,a) = \Phi(k,a)\left[1 - \chi_\Phi\left(\frac{k}{k_{\mathrm{rel}}}\right)^\alpha\right]. \tag{1}$$

Large-scale modes, which dominate the collapse of massive halos, retain a high degree of temporal coherence. As a result, clusters form earlier and with deeper potential wells than in the standard cosmological model. Small-scale modes, which would normally contribute to the steepening of the inner density profile, are suppressed, leading to smoother central regions.

The virial temperature of the intracluster medium becomes

$$T_{\mathrm{vir,RDCC}} = T_{\mathrm{vir}} \left[1 - \chi_T \left(\frac{k(M)}{k_{\mathrm{rel}}} \right)^{\alpha} \right], \tag{2}$$

reflecting the modified collapse dynamics.

3 Pressure Profiles and the Compton-y Parameter

The tSZ effect is quantified by the Compton-y parameter,

$$y = \frac{\sigma_T}{m_e c^2} \int P_e(r)\, dl, \tag{3}$$

where $P_e(r)$ is the electron pressure.

In RDCC, the pressure profile is modified by the relaxation mechanism. The hydrostatic equilibrium condition,

$$\frac{1}{\rho}\frac{dP}{dr} = -\frac{d\Phi_{\mathrm{RDCC}}}{dr}, \tag{4}$$

yields a pressure distribution that is smoother in the core and more extended in the outskirts.

The RDCC pressure profile can be approximated as

$$P_{\mathrm{RDCC}}(r) = P(r) \left[1 - \chi_P \left(\frac{1}{k_{\mathrm{rel}} r} \right)^{\alpha} \right], \tag{5}$$

where $P(r)$ is the standard pressure profile.

This modification enhances the tSZ signal in the outskirts while reducing it in the core.

4 Integrated tSZ Signal and Scaling Relations

The integrated tSZ signal is given by

$$Y = \int y\, d\Omega. \tag{6}$$

In RDCC, the integrated signal becomes

$$Y_{\mathrm{RDCC}} = Y \left[1 - \chi_Y \left(\frac{k(M)}{k_{\mathrm{rel}}} \right)^{\alpha} \right]. \tag{7}$$

This modification alters the scaling relation between Y and halo mass. Massive clusters, whose formation is accelerated by the coherence of large-scale modes, exhibit enhanced tSZ signals. Low-mass clusters, whose collapse is delayed by the relaxation mechanism, show suppressed tSZ signals.

These deviations from self-similar scaling provide a direct observational probe of the relaxation scale.

5 Observational Signatures

The RDCC modifications to the tSZ effect manifest in several measurable ways. Clusters exhibit smoother pressure profiles, enhanced tSZ signals in their outskirts, and deviations from standard scaling relations. The tSZ power spectrum is similarly affected, with a suppression of small-scale power and an enhancement of large-scale coherence.

These signatures provide a direct observational window into the relaxation scale and the temporal structure of the gravitational field in RDCC.

6 Conclusion

The thermal Sunyaev–Zel'dovich effect in the Relaxation-Driven Cyclic Cosmology reflects the scale-dependent evolution of the gravitational potential. The relaxation mechanism modifies the formation history of clusters, the thermodynamic structure of their gas, and the pressure distribution that determines the tSZ signal. These effects produce distinctive signatures in the Compton-y parameter, the integrated tSZ signal, and the tSZ power spectrum. The present Companion establishes the theoretical foundation for future studies of cluster astrophysics, CMB secondary anisotropies, and the interplay between baryons and dark matter in RDCC.

References

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